# 12 PHYSICS
# PROJECTILE MOTION INVESTIGATION

*Purpose*

To investigate projectile motion using a suitable Java applet.

*Internet Site*

http://physics.uwstout.edu/physapplets

Select "Projectile Motion".

**Investigations**

Manipulate the input data to investigate the following situations.

In each investigation, you need to do the following.

- Determine the ***dependent, independent and controlled variables***. List them.
- Record the data in a suitable table.
- Use your ***calculator*** to graph the data.
- Graph the data ***manually*** on graph paper.
- Determine the relationship from the shape of the graph.

1. What is the relationship between the ***angle of projection*** and ***range***? (Assume no air resistance.)
2. What is the relationship between the ***launch velocity*** and ***range***? (Assume no air resistance.)
3. What is the relationship between the ***mass of an object*** and the ***range***, allowing for air resistance?

*Write Up*

For each of the investigations, neatly and clearly record your data in tables.

Draw separate graphs on graph paper.

State clearly your conclusion for each investigation, which must be based on your data and graph.